

PFEM Benchmark Catalogue

From Deterministic Sweeps to Probabilistic Analysis

Naty Manque & Naeem Zainuddin

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Where We Are

Done

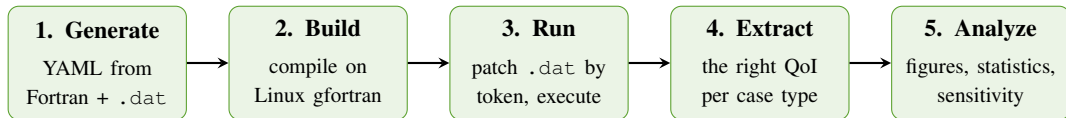
- ✓ **Foundation:** all 87 PFEM cases build, run and verified on Linux
- ✓ **Deterministic sweeps** + 3D post-processing (example: p6.12 slope)
- ✓ **Phase 1:** universal Quantity-of-Interest extraction + Monte Carlo
- ✓ **Phase 2:** Latin Hypercube, correlated inputs, sensitivity tornado

Next

- ▶ Phase 3: pluggable runner for non-PFEM codes
- ▶ Phase 4: mesh-refinement convergence studies
- ▶ Regression testing vs reference outputs

Handover: complete documentation set and a 3-minute reproducible test, all on GitHub (final slides).

How It Works — One Pipeline, Five Stages



Key idea

- ▶ YAML is the bridge: parameters patched by **token index**
- ▶ The Fortran **source is never edited**
- ▶ One driver runs all **87** cases

Same core, four modes

- ▶ Deterministic sweep
- ▶ Monte Carlo / Latin Hypercube
- ▶ Correlated sampling
- ▶ One-at-a-time sensitivity

Example case: 3D slope stability (p6.12) — Mohr-Coulomb with strength reduction

p6.12 — 3D Slope Stability (Mohr-Coulomb)

Physical Problem

- ▶ 3D embankment slope, elastic-perfectly plastic
- ▶ Mohr-Coulomb failure criterion with 5 material zones
- ▶ Hex8 mesh, 20-point Gaussian integration

Strength Reduction Factor (SRF)

Shear strength is scaled down uniformly at each step:

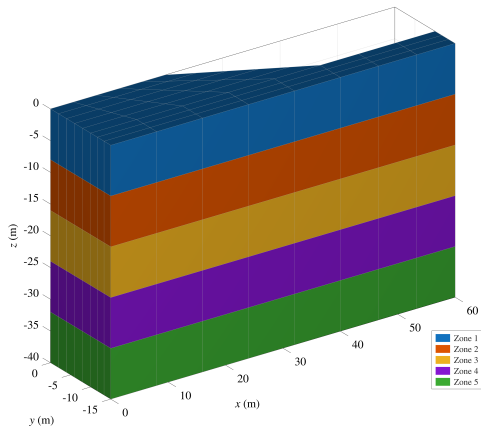
$$\hat{c} = \frac{c}{\text{SRF}}, \quad \tan \hat{\phi} = \frac{\tan \phi}{\text{SRF}}$$

Failure = last SRF at which Newton-Raphson

Default Parameters

Parameter	Value
Cohesion c	60 kPa
Friction ϕ	20°
Dilation ψ	0°
Young's E	10 ⁵ Pa
Poisson ν	0.3
SRF range	1.0 – 1.60
Iteration limit	1000

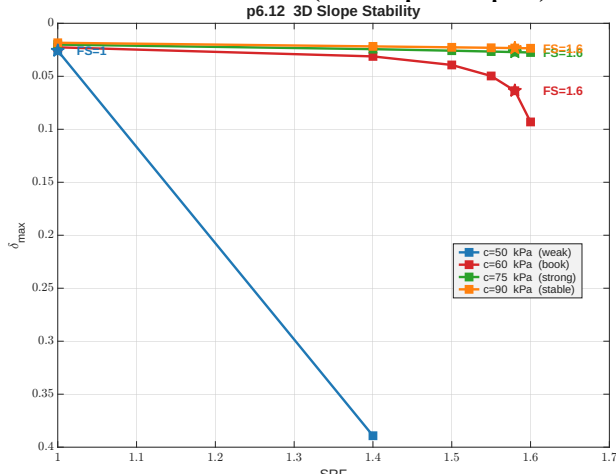
p6.12 — Material Zones



5 horizontal soil layers

- ▶ Each zone has its own cohesion c and friction angle ϕ
- ▶ Geometry is **identical** across all sweep scenarios
- ▶ Only the base cohesion value changes; zone proportions are preserved
- ▶ Slope face visible on the right side

p6.12 — SRF Curves (Sweep Output)



What this shows

- ▶ X-axis: Strength Reduction Factor applied to c and ϕ
- ▶ Y-axis: maximum displacement $\delta_{\max}$
- ▶ $c=50$ kPa: slope fails at $SRF \approx 1.4$ — displacement diverges
- ▶ $c=60$ kPa (book): $FS \approx 1.6$
- ▶ $c=75$, $c=90$: stable — no divergence

Key: diverging $\delta_{\max}$ signals slope failure.

p6.12 — Verification vs. Book (Figs. 6.54–6.55)

SRF vs. Max Displacement ($c=60$, $\phi=20^\circ$)

SRF	$\delta_{\max}$	Iterations
1.00	0.02266	16
1.40	0.03123	77
1.50	0.03930	208
1.55	0.04971	380
1.58	0.06363	703
1.60	0.09308	1000

Failure at $\text{SRF} \approx 1.58\text{--}1.60$

Verification

- ✓ SRF–displacement curve matches book Fig. 6.54
- ✓ Failure $\text{SRF} \approx 1.58\text{--}1.60$ agrees with reference
- ✓ 3D slip surface visible in deformed mesh (Fig. 6.55)
- ✓ Iteration count growth consistent with limit-state analysis

Phase 1 — From One Answer to a Distribution

The problem

- ▶ A deterministic run gives *one* number for *one* set of inputs
- ▶ Real material properties are uncertain (scatter, test variability)
- ▶ The old extractor only understood slope-stability output

What was added

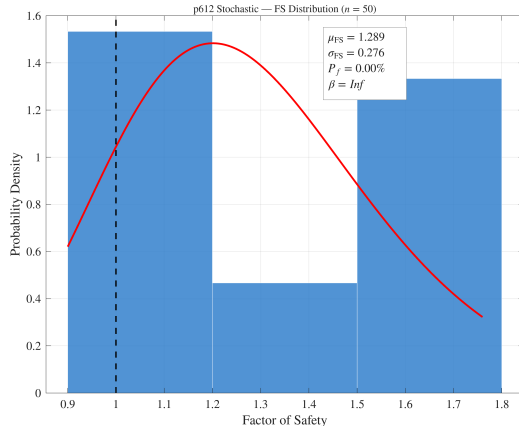
- ✓ A **universal QoI extractor**: pulls the right physical answer for **all 87** cases (8 case types)
- ✓ **Monte Carlo**: sample inputs from distributions, get a distribution of answers

Case type → Quantity of Interest

Case type	QoI
Slope stability	Factor of Safety
Plasticity	Limit load
Elastic	Max displacement
Seepage	Max head
Consolidation	Degree of consol.
Eigenvalue	ω^2 (and f_1)
Dynamics	Peak displacement
Thermal	Max temperature

87 / 87 cases return a meaningful QoI from their default run.

Phase 1 — Result: Slope Reliability (p6.12)



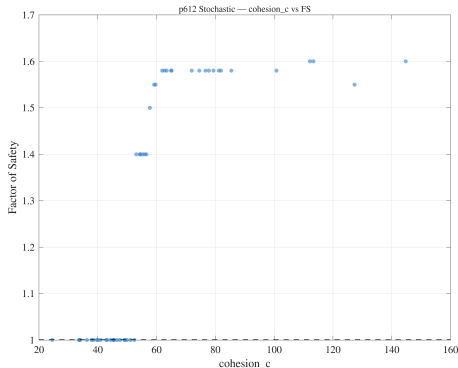
Cohesion sampled as lognormal

- ▶ Each Monte Carlo sample runs the full 3D Fortran solver
- ▶ Output: a **distribution** of Factor of Safety, not a single value
- ▶ Gives **probability of failure** $P(FS < 1)$ and reliability index β

This is the language a geotechnical reviewer actually wants: not "FS = 1.58" but "probability of failure = X%".

Phase 2 — Latin Hypercube: Same Cost, Better Coverage

Variance reduction vs plain Monte Carlo



Samples	LHS std	IID std
25	0.75	4.92 (6.5×)
50	0.54	3.42
100	0.20	2.33
200	0.11	1.58 (14×)

- ▶ Same number of expensive Fortran runs
- ▶ 6–14× tighter estimate of the mean
- ▶ Scatter (left) shows FS rising smoothly with cohesion — physically correct

Phase 2 — Correlated Inputs (Iman–Conover)

Why

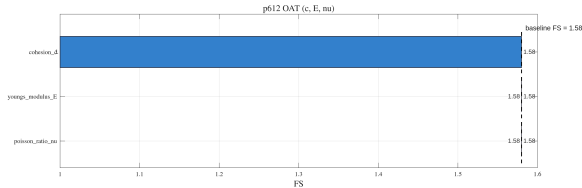
- ▶ In real soils, cohesion c and friction ϕ are correlated (typically negatively)
- ▶ Sampling them independently is unphysical
- ▶ Iman–Conover induces a **target correlation** while keeping each input's distribution exact

Reproduced target correlations ($n=500$)

Target ρ	Observed ρ
−0.70	−0.654
−0.30	−0.275
+0.00	−0.002
+0.50	+0.494

Targets in $[-0.7, +0.5]$ reproduced within 5%, marginals preserved exactly.

Phase 2 — Sensitivity: Which Input Matters?



One-at-a-time tornado (p6.12)

- Each input varied $\pm 1\sigma$; bar length = effect on FS
- **Cohesion** dominates (spread 0.58)
- E and ν have *zero* effect on FS

Exactly the textbook expectation: in strength-reduction analysis only strength parameters move the safety factor; stiffness only affects displacements. The tool **recovers the physics.**

Why We Trust It — Validation

Case	Computed	Reference	Match
p6.12 slope FS ($c=60, \phi=0$)	1.58	Book Fig. 6.54	✓
p6.1 limit load ($\sigma_y=100$)	515	Prandtl $(2+\pi)\sigma_y = 514$	✓ 0.2%
p10.1 beam ω^2	0.00382	Analytic cantilever 0.00402	✓
LHS variance reduction	6.5–14×	McKay et al. (1979)	✓
Correlation reproduction	within 5%	Iman–Conover (1982)	✓

- Results match the **textbook** and **analytical** solutions, not just "it runs"
- A single 3-minute script (`test_phase2_multi_case.m`) reproduces this evidence on demand

Handover — Anyone Can Pick This Up

Everything is documented and on GitHub

- ▶ ARCHITECTURE.md — end-to-end flow, how every file connects, the call graph, every problem solved
- ▶ HANDOVER.md — fresh-machine setup, status, gotchas
- ▶ GUIDE.md / matlab/README.md — usage and function reference
- ▶ PROGRESS.md — this validation evidence in full

Reproducible from a cold start

- ▶ A literal **"zero to first result"** runbook: clone → build → run → figure
- ▶ 3-minute test script reproduces all validation
- ▶ Documented patches make the textbook Fortran compile on Linux

The only thing not in the repo is the licensed PFEM source — but the exact patches to rebuild it are.

Conclusions & Next Steps

What was achieved

- ✓ 87 PFEM benchmarks verified on Linux (token-based YAML, zero source changes)
- ✓ Universal QoI extraction across all 8 case types
- ✓ Monte Carlo + Latin Hypercube + correlated inputs
- ✓ One-at-a-time sensitivity (tornado) recovering the physics
- ✓ Validated vs textbook and analytical solutions
- ✓ Full handover documentation on GitHub

Next steps

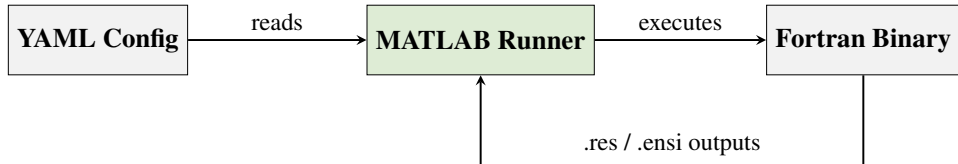
- ▶ Phase 3: pluggable runner for non-PFEM codes
- ▶ Phase 4: mesh-resolution convergence sweeps (*nxe*, *nye*)
- ▶ Automated regression testing against reference outputs
- ▶ User-selectable probe node for boundary-bound QoIs

Thank You!

Questions?

Backup Slides

Pipeline Architecture



YAML acts as the bridge between rigid Fortran and flexible MATLAB.

Per-run steps

1. Locate binary + template .dat
2. Copy to an isolated sandbox folder
3. Inject override values as positional tokens
4. Execute; parse output files

Example tunable tokens

- ▶ `youngs_modulus_E` → replaces value in .dat
- ▶ `cohesion_c, friction_phi` → MC sweep
- ▶ One YAML .. many scenarios — no

Sweep GUI — pfem_sweep_gui

A single MATLAB interface for all 87
benchmark cases:

- ▶ Browse cases by chapter; inspect tunables with suggested ranges
- ▶ Define sweeps on a **log or linear** grid
- ▶ Execute variants in isolated sandboxes
- ▶ Auto-generate summary figures (mesh, displacement, SRF, 3D EnSight)

1. Select Case



2. Browse Tunables



3. Set Sweep Grid



4. Run All Variants



5. View Summary Plots

Cases

chap06/p612
chap05/p61_3

+ Add YAML(s)

- Remove Selected

Run

Run All

Stop

Done (5/5 OK)

Tunable Parameters — enable, enter values, see suggested ranges

	Parameter	Values (comma-separated)	Suggested Range	Chapters
1	☒ cohesion_c	0.6 60 6000	[0.6, 6000]	6
2	☒ convergence_tolerance	1e-12 3.162e-07 0.1	[1e-12, 0.1]	6
3	☒ dilation_angle_psi	0.045 1.423 45	[0, 45]	6
4	☒ friction_angle_phi	0.045 1.423 45	[0, 45]	6
5	☒ iteration_limit	10 316 1e+04	[10, 1e4]	6
6	☒ nels_or_nxe			6, 5
7	☒ np_types_or_nye			6, 5
8	☒ poisson_ratio_nu	0.00049 0.0155 0.49	[0, 0.49]	6, 5
9	☒ unit_weight_gamma	0.1 3.162 100	[0, 100]	6
10	☒ youngs_modulus_E	1000 1e+05 1e+07	[1000, 1e7]	6, 5

Mode: Lockstep (same-length arrays)

Fill Ranges

-

3

+

Preview Scenarios

Log

```

Step 1: Add YAML cases. Step 2: Configure parameters. Step 3: Run All.
=== Run Start: 2 case(s) x 3 scenario(s) = 6 total ===
--- Case: p612 (chap06) ---
(1/6) p612 | c=0.6 tol=1e-12 psi=0.045 phi=0.045 lim=10 nu=0.00049 Um=0.1 E=1000
OK - 2 file(s) max|u| = 2.137e-01 t = 8.87 s
(2/6) p612 | c=60 tol=3.16e-7 psi=1.423 phi=1.423 lim=316 nu=0.0155 Um=3.162 E=1e5
OK - 2 file(s) max|u| = 4.075e-02 t = 5.00 s
(3/6) p612 | c=6000 tol=0.1 psi=45 phi=45 lim=1e4 nu=0.49 Um=100 E=1e7
OK - 2 file(s) max|u| = 1.718e-02 t = 0.76 s
--- Case: p61_3 (chap05) ---
(4/6) p61_3 | c=0.6 tol=1e-12 psi=0.045 phi=0.045 lim=10 nu=0.00049 Um=0.1 E=1000
OK - 4 file(s) max|u| = 1.000e-05 t = 0.06 s
(5/6) p61_3 | c=60 tol=3.16e-7 psi=1.423 phi=1.423 lim=316 nu=0.0155 Um=3.162 E=1e5
OK - 4 file(s) max|u| = 1.000e-05 t = 0.03 s
(6/6) p61_3 | c=6000 tol=0.1 psi=45 phi=45 lim=1e4 nu=0.49 Um=100 E=1e7
OK - 4 file(s) max|u| = 1.000e-05 t = 0.04 s
=== Run complete ===

```

Results

	Case	Scenario	Status	max u	Time (s)	Run Directory
1	☐ p612	c=0.6 tol=1e-12 psi=0.045 phi=0.045 lim=10	OK	2.137e-01	0.87	/home/raeeem/projects/fem-benchmarks/runs/chap06/p612/c_0.6_tol_
2	☐ p612	c=60 tol=3.16e-7 psi=1.423 phi=1.423 lim=316	OK	4.075e-02	5.00	/home/raeeem/projects/fem-benchmarks/runs/chap06/p612/c_60_tol_
3	☐ p612	c=6000 tol=0.1 psi=45 phi=45 lim=1e4 nu=0.49	OK	1.718e-02	0.76	/home/raeeem/projects/fem-benchmarks/runs/chap06/p612/c_6000_tol_
4	☐ p61_3	c=0.6 tol=1e-12 psi=0.045 phi=0.045 lim=10	OK	1.000e-05	0.06	/home/raeeem/projects/fem-benchmarks/runs/chap05/p61_3/c_0.6_tol_
5	☐ p61_3	c=60 tol=3.16e-7 psi=1.423 phi=1.423 lim=316	OK	1.000e-05	0.03	/home/raeeem/projects/fem-benchmarks/runs/chap05/p61_3/c_60_tol_

Open Figures

Show Comparison

Clear Results

Problem Statement

Developing benchmark Fortran codes (PFEM, 5th ed.) by automating YAML metadata generation to tokenise `.dat` inputs.

Key Challenges

- ▶ **Legacy Input:** Unstructured `.dat` files.
- ▶ **Opaque Parsing:** The `read10` subroutine hides logic.
- ▶ **Constraint:** Source code must remain untouched.

Objectives

- ▶ **Reproducibility:** Run any case reliably.
- ▶ **Interoperability:** Drive execution via MATLAB.
- ▶ **Searchability:** Consistent YAML metadata.

Path to Completion

To transition from prototype to a **research-ready benchmark repository**:

1. **Standardise YAML Schema (All Chapters)**

- ▶ Uniform metadata keys and consistent tunable definitions.

2. **Define Scope of Tunables**

- ▶ **Phase 1:** Material props (E, ν, k), loads.
- ▶ **Phase 2:** Mesh resolution (n_{xe}, n_{ye}).

3. **Implement Validation Layer**

- ▶ Health checks and regression testing against reference outputs.

p5.1 — Problem Setup (Linear Elasticity 2D)

Physical Problem

- ▶ Plane-stress elastic solid under point loads
- ▶ Mesh: 3×2 bilinear quadrilateral elements (12 nodes, 6 elements)
- ▶ Fixed bottom edge; two point loads applied at top

Tunable Parameters

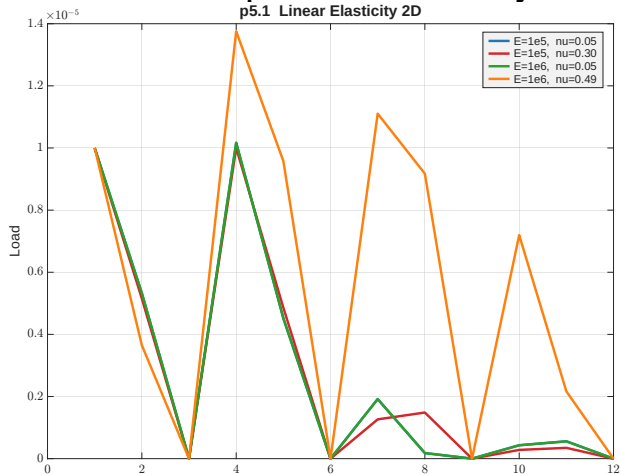
Parameter	Symbol	Range
Young's modulus	E	10^4 – 10^8
Poisson's ratio	ν	5×10^{-4} – 0.49

YAML snippet

```
id: pfem5_ch05_p51_p51_3
title: PFEM p51 -- case p51_3
purpose: Linear Elasticity 2D

tunable_parameters:
- name: youngs_modulus_E
  current_value: '1.0e6'
  suggested_range: [1e4, 1e9]
- name: poisson_ratio_nu
  current_value: '0.3'
  suggested_range: [0.0, 0.49]
```

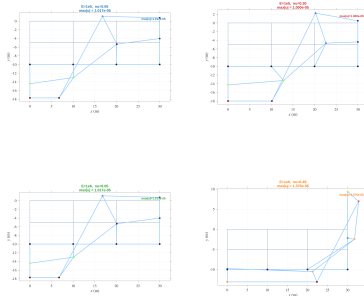
p5.1 — Load–Displacement Overlay



What this shows

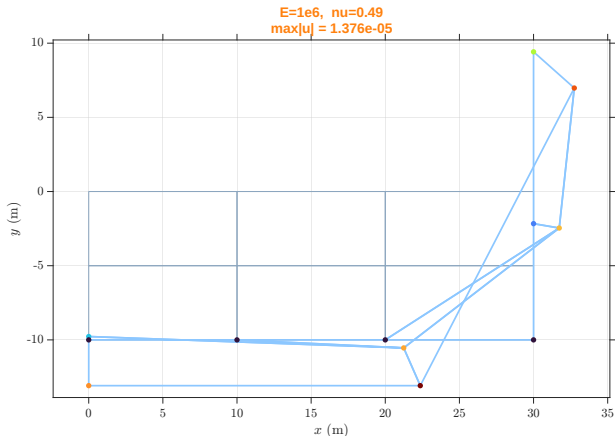
- ▶ 4 scenarios: E and ν varied on a log grid
- ▶ $\nu=0.49$: near-incompressible, strong Poisson effect
- ▶ Low ν scenarios cluster together — stiffness dominates

p5.1 — Deformed Shapes (All Scenarios)



- Poisson ratio dominates the deformed shape; $\nu=0.49$ shows strong lateral bulging
- Grey = undeformed mesh, blue = amplified deformed shape

p5.1 — Zoom: $E=10^6$, $\nu=0.49$ (Near-Incompressible)



Observation

- ▶ Near-incompressible material ($\nu \rightarrow 0.5$)
- ▶ Large lateral expansion dominates the deformed shape
- ▶ Vertical load produces significant horizontal displacement

p5.1 — Verification vs. Book (Table 5.3)

Nodal Displacements

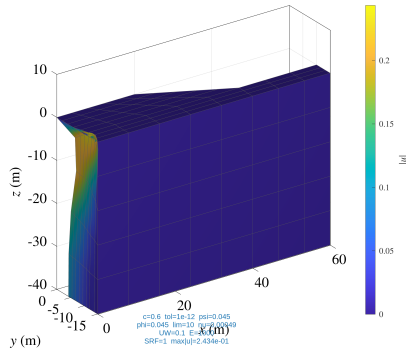
Default: $E = 10^6$, $\nu = 0.3$, plane stress

Node	u_x	u_y
1	0.000	-1.000×10^{-5}
4	9.446×10^{-6}	-1.000×10^{-5}
5	8.592×10^{-6}	-4.235×10^{-6}
10	0.000	$+7.196 \times 10^{-6}$

Verification

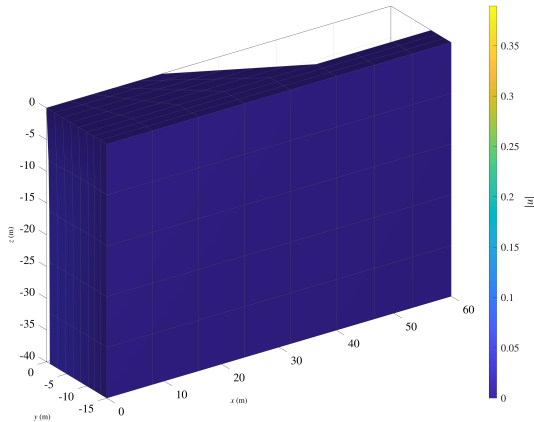
- ✓ Displacements match book
Table 5.3 exactly
- ✓ Integration point stresses match
Table 5.3
- ✓ Sweep shows correct $1/E$ scaling

p6.12 — 3D Deformed Mesh at Failure (All Scenarios)



- Each panel: deformed mesh at the highest converged SRF step; colour = $|u|$
- Top-left ($c=50$): severe slope failure. Bottom-right ($c=90$): minimal deformation

p6.12 — $c = 50$ kPa: Slope Failure

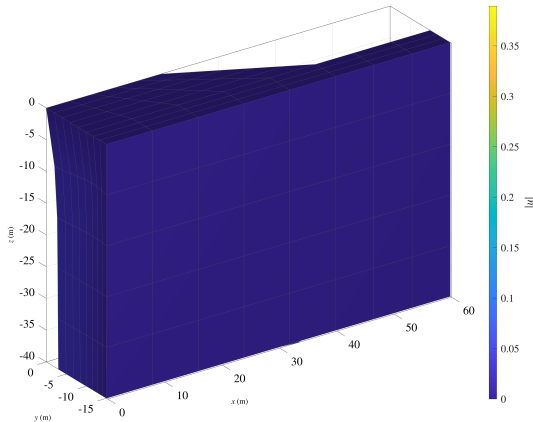


Weak soil scenario

- ▶ Newton-Raphson diverges at $\text{SRF} \approx 1.4$
- ▶ Clear circular slip surface visible at the slope crest
- ▶ Large horizontal displacement at the toe
- ▶ Colourmap confirms displacement localisation along failure plane

This scenario demonstrates **how the tool detects failure** automatically.

p6.12 — $c = 60$ kPa: Book Default

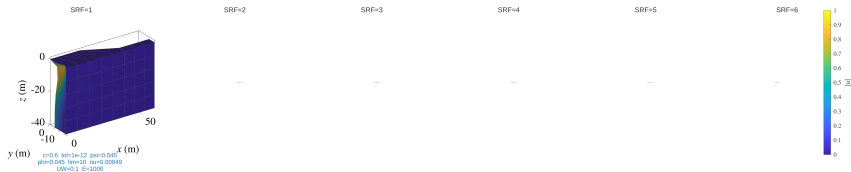


Reference scenario

- ▶ Incipient failure at $\text{SRF} \approx 1.6$ (iteration limit reached)
- ▶ Deformed shape matches book Fig. 6.55 (PFEM 5th ed.)
- ▶ Moderate displacement — slope is on the verge of collapse

Validates our pipeline against the published reference solution.

p6.12 — SRF Progression (Deformation Sequence)



- Rows = scenarios ($c=50 \rightarrow 90$ kPa). Columns = SRF steps 1.0–1.6
- Weak soil: progressive failure visible across columns. Strong soil: stable throughout